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To cite this article: Bowen Yan & Yefan Teng (07 Feb 2025): The double-edged sword effect of artificial intelligence awareness on organisational citizenship behaviour: a study based on knowledge workers, Behaviour & Information Technology, DOI: [10.1080/0144929X.2025.2462267](https://doi.org/10.1080/0144929X.2025.2462267)

To link to this article: <https://doi.org/10.1080/0144929X.2025.2462267>



Published online: 07 Feb 2025.



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The double-edged sword effect of artificial intelligence awareness on organisational citizenship behaviour: a study based on knowledge workers

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ABSTRACT

Artificial Intelligence (AI) has been developing explosively in recent years, and in this context, knowledge workers are increasingly understanding and using AI, and AI awareness influences their behaviours. Based on self-efficacy theory, this study constructed a dual-path model aiming to reveal the impact of knowledge workers' AI awareness on their organisational citizenship behaviours. The study collected multi-source data from China ($n = 461$) to construct a structural equation model, and the results indicated that AI awareness has a double-edged sword effect on organisational citizenship behaviours of knowledge workers, which are positive effect by promoting performance self-efficacy and negative effect by inhibiting creative self-efficacy, respectively. In the process of promoting organisational citizenship behaviours, organisational innovation climate plays a positive moderating role. This study also discussed the implications of the findings for both theory and practice.

ARTICLE HISTORY

Received 9 July 2024

Accepted 29 January 2025

KEYWORDS

AI awareness; self-efficacy; organisational citizenship behaviour; organisational innovation climate; knowledge worker

1. Introduction

Artificial Intelligence (AI) is considered the driving force that is transforming the existing production model of business and propelling society into the Industry 4.0 era (Shankar et al. 2024). At the macro level, this technology has brought about a radical change in the business model of modern society, with AI technology increasing the profitability of companies by improving their production techniques (Sahoo et al. 2024). At the micro level, during the alternation between traditional and new technologies in which AI technology is gradually becoming a major productive force, there is uncertainty about where AI technology will lead humanity, and this has had a variety of impacts on individual perceptions and behaviours (Prentice, Wong, and Lin 2023). The impact of AI technology on individual behaviour acts firstly on the individual's perception of AI, and in academia this perception is referred to as AI awareness (AIA) (Zhang et al. 2022). Within the workplace, AI awareness is often thought to have a bidirectional impact on employee behaviour. On the one hand, AI technology helps to reduce the workload of employees and increase their efficiency (Prentice and Nguyen 2020). On the other hand, employees may perceive themselves as dispensable in the work environment due to the promotion of AI technology, leading to a range of negative perceptions, such as job

insecurity and emotional exhaustion, which in turn lead to the implementation of a variety of negative behaviours (Lingmont and Alexiou 2020; Teng et al. 2023). Existing studies have confirmed the double-edged sword effect of AI awareness on employees' turnover intention, innovative behaviour, and performance (Leong et al. 2019; Liang et al. 2022).

Although academics have begun to explore the impact of AI on employees' in-role behaviours, and in particular, some scholars have noted that AI awareness significantly affects employees' positive work behaviours (Cheng, Lin, and Kong 2023), however, the link between AI awareness and employees' positive extra-role behaviours has still not been established, and there is a clear gap in this research field. Positive extra-role behaviours by employees are behaviours that benefit the organisation but are beyond their responsibilities and will not be explicitly rewarded or reciprocated by the organisation, yet often have a significant positive impact on organisational development (Liu, Ko, and Chapleo 2017). Organisational citizenship behaviour (OCB) is considered one of the important extra-role behaviours implemented by employees that must be attended to in the development process of an organisation (Ma, Teng, and Yan 2024). Unfortunately, the existing literature does not establish any link between AI awareness on any employee extra-role behaviours, including

OCB, which limits the scholars' understanding of the mechanisms by which AI awareness affects an important individual behaviour of employees. Moreover, most of the existing literature focuses on the study of the impact of AI awareness on employees' individual behaviours in the service industry, such as the hospitality and tourism industries (Errichiello and Drago 2023; Liang et al. 2022; Teng et al. 2023). Compared with other types of employees, knowledge workers are more sensitive to technological development, are more capable of learning, more willing to proactively try new technologies, and have richer scenarios of applying AI in the workplace (Drucker 1999), therefore, this study argues that the impact of AI awareness on the behaviour of knowledge workers is much more worthy of attention. However, the results of the impact of knowledge workers' AI awareness have indeed been neglected in previous studies (Yao et al. 2023), and no literature has yet examined it. In the era of the knowledge economy with increasingly fierce market competition, organisations require knowledge workers to adjust and improve their work at any time to cope with the unpredictable external market environment, so relying only on rules and regulations to manage the in-role behaviours of knowledge workers is not enough to fully cope with the development of the organisation's needs, and the OCB of knowledge workers is particularly important for the organisation (Blatt 2008). Therefore, the attempt to explore the influence of AI awareness on knowledge workers' OCB is both theoretically important and practically valuable.

Some previous studies have explored the impact of AI on employee behaviour based on the resource conservation theory, and AI awareness is considered a challenging stressor and a hindering stressor to confirm the double-edged sword effect of AI on employee behaviour, and while AI on one hand positively affects in the areas of work autonomy, Job crafting and motivation, on the other hand, AI has a negative impact in terms of employees' emotional exhaustion and job insecurity (Cheng, Lin, and Kong 2023; Liang et al. 2022; Zhang et al. 2023). Here, we make the argument that the use of that single theory may lead to some limitations in the academic exploration of the influence pathways between AI and employee behaviour. Self-efficacy theory (SET) emphasises the direct influence of people's cognitive judgments of changing environments on their behaviours, which can explain the cognitive logic behind individuals' complex social behaviours (Ajzen 2002), at the same time, stimulus-organism-response (SOR) model also has good applicability in studying employee behaviour (Tang, Warkentin, and Wu 2019). Therefore, we drew on the perspective of the SOR

model and introduced SET to address the above research gaps, using creative self-efficacy (CSE) and performance self-efficacy (PSE) as mediating variables, in order to explore the double-edged sword effect of AI awareness on knowledge workers' OCB due to its facilitating effect on their PSE as well as its inhibiting effect on their CSE. In addition, it has been confirmed that organisational innovation climate (OIC) enhances employees' sense of organisational support, contributes to the rapid diffusion of new technologies in the organisation (Ali, Lei, and Wei 2018), alleviates employees' insecurity about the unknown caused by technological change, and has a promoting effect on employees' innovative behaviour (Shanker et al. 2017). Specifically, a higher level of OIC encourages employees to adopt new technologies and implement innovative behaviours, employees' PSE increases as a result of increased efficiency (Jun Choi and Park 2014; Xu et al. 2023), and the innovative behaviour of applying new technologies itself reduces the negative impact of AI on employees' CSE (Puente-Diaz et al. 2020). Therefore, we believe that the OIC plays a crucial role in the process of AI awareness influencing employees' two types of self-efficacy. In summary, this study explores the underlying mechanisms and boundary conditions of the impact of knowledge workers' AI awareness on OCB by constructing a moderated double-edged sword mediation model.

The theoretical contributions of this study include: firstly, it enriches the literature on the impact of AI awareness on employee behaviour, particularly focusing on knowledge workers; secondly, this study introduces the SET that has never been applied in this field to explain how AI awareness has an inhibitory and promoting effect on OCB, helping the scholars to open up the 'black box' of the double-edged sword impact of AI awareness on employees' OCB, and expanding the application of SOR model; thirdly, unlike previous studies that broaden the boundary conditions of the impact of AI on employee behaviour from an individual perspective, this study explores from the organisational level perspective of OIC, providing new ideas for subsequent research. The research results provide practical implications for managers to correctly guide knowledge workers to consider AI and benefit organisations by increasing the OIC.

2. Theory and hypotheses development

2.1. Self-efficacy theory (SET)

SET has been widely used in research to explain how the external environment affects employee performance (Bachrach et al. 2023). Self-efficacy refers to people's

confidence and expectations about their ability to accomplish a goal in a given situation, emphasising subjective evaluations of their abilities (Bandura 1977). Individuals' cognition is affected by changes in the environment, which in turn leads to significant changes in behaviour, as shown by the fact that individuals with high self-efficacy put more effort into achieving an individual or group goal, while those with low self-efficacy are more likely to avoid or withdraw from it (Maran et al. 2022). Scholars usually categorise self-efficacy into two aspects: PSE and CSE (Tang and Wei 2022). PSE comes from employees' perceptions of their ability to accomplish work goals, in this study, knowledge workers with higher AI awareness can better complete daily complicated paperwork through ChatGPT and other methods to help them improve their work efficiency, and in this process, employees' AI awareness positively affects behaviour by stimulating positive perceptions (Ali et al. 2023). CSE comes from employees' beliefs about their abilities, in this study, AI technology increased employees' job insecurity due to the substitution of AI technology for knowledge workers' skills, in today's rapid technological change employees will doubt whether their abilities are able to match the ever-changing market of labour demand, in this process, AI awareness stimulates such negative perceptions and then inhibits the positive behaviours. Therefore, we believe it is appropriate to explore the impact of employees' AI awareness on OCB based on SET.

2.2. Stimulus-organism-response (SOR) model

SOR model suggests that external stimuli affect behaviour by influencing the emotions or psychology of an individual (Mehrabian and Russell 1974). The model consists of three components. Stimulus as the first component can be various external events, such as actual events, the behaviour of others, social attitudes; organism as the second component can be the individual's emotional responses after perceiving external events, such as emotional changes, subjective perception changes, mental state changes; response as the third component is the final externalised manifestation of the changes in the second component due to the first component, which can be a certain tendency or actual behavioural manifestation (Jang and Namkung 2009; Suh and Prophet 2018). SOR model was initially used to solve environment-related psychological problems, and in recent years, it has also been widely used in the field of organisational behaviour (Tang, Warkentin, and Wu 2019). In this study, AI awareness maps the external stimulus of AI technology to knowledge-based employees, which represents the external stimulus

affecting self-efficacy (i.e. the organism). Ultimately, employees' OCB changes as a result (i.e. the response). Therefore, it is appropriate for this study to construct a theoretical framework based on SOR model.

2.3. AI Awareness and OCB

In recent years, AI, an emerging technology, has realised a huge breakthrough from the laboratory to the market, and it is an important booster to promote enterprises to enter 'Industry 4.0' (Obreja, Rughiniş, and Rosner 2024; Shankar et al. 2024). The use scenarios of AI (such as ChatGPT, Google Bard, BoardMix) within the work scenarios of knowledge workers are becoming more and more widespread, and the use of these AI programmes in work scenarios has helped to enhance convenience, which in turn improves productivity and helps users to keep track of the latest knowledge and information promptly (Wang, Gao, and Agarwal 2023). However, people have also become worried about the potential pitfalls of AI development, such as the possibility of replacing a large number of jobs, leading to a decrease in interpersonal interactions, privacy issues, and unethical behaviours (Teng et al. 2023; Willems et al. 2023). AI awareness, which refers to an individual's subjective assessment of the technology, is an important antecedent variable often used in studies examining the impact of AI as a technology on employee behaviour (Khaliq et al. 2022). Existing studies have examined the positive and negative effects of AI awareness, for example, it has been demonstrated in the literature that while AI awareness increases performance (Kong et al. 2021), it also leads to employee burnout, emotional exhaustion, and increased turnover intentions (Henkel et al. 2020; Li, Bonn, and Ye 2019; Tojib et al. 2023; Yang, Mei, and Zheng 2023), and even breeds unethical behaviour (He et al. 2024). Based on this, we believe that the research direction of simultaneously exploring both positive and negative aspects of the impact that AI awareness may have on employee behaviour is of significant value.

Organizational Citizenship Behavior (OCB) refers to employee-initiated behaviours that go above and beyond their job tasks and responsibilities (Dalal 2005). OCB is not bound by rules within the organisation and is not usually rewarded in any form by the organisation but has a positive contribution to organisational performance (Alt and Spitzack 2016). Some studies have explored the antecedent variables and influencing mechanisms of employees' OCB from different directions, among which a study based on SET pointed out that self-efficacy is an important

antecedent variable that affects OCB, and that if employees are compared to batteries, strong self-efficacy of employees means that they are fully charged, and then employees will be more willing to apply their energy to implement more altruistic behaviours (Choong et al. 2020). With the development of science, AI technology has had a great impact on knowledge workers. From a positive point of view, the improvement of work efficiency by AI increases the possibility of employees completing complex work tasks, which has a positive effect on their self-efficacy, and thus increases their OCB (Alessandri et al. 2021). From a negative point of view, the ability of AI to think equally with humanity makes it easier to replace the job skills of knowledge workers, leading to a sense of crisis by science and technology and then doubt their ability, which has a depressing effect on their self-efficacy, and their motivation to protect their resources will correspondingly reduce the OCB (Ozyilmaz, Erdogan, and Karaeminogullari 2018). Therefore, this study will explore the double-edged sword effect of knowledge workers' AI awareness based on SET in the framework of SOR model, aiming to enrich the related research on the impact of AI on knowledge workers.

2.4. The mediating role of PSE

As mentioned earlier, the application of AI in the working scenario of knowledge workers has become very common, and AI software and programmes have even become an important tool for assisting knowledge workers in completing their work tasks: in addition to being able to help them complete their basic paperwork, more specialised AI software has even been able to partially complete the employees' key tasks (Cramarenco, Burcă-Voicu, and Dabija 2023; LazaroIU and Rogalska 2023). In this process, employees only need to review and optimise the results provided by AI, which greatly reduces the heaviness of their daily work (Peters et al. 2024). PSE is a self-perception of the employee's ability to accomplish work tasks and solve work problems (Shea and Howell 2000). Compared with the grassroots employees in labour-intensive industries, knowledge workers have a higher degree of demand for technology and information, when knowledge workers have a strong sense of emerging technology, they are more willing to take the initiative to learn and try (Shujahat et al. 2019), which is consistent with the reality, some office type of AI software as soon as it comes to the market quickly caused dramatic changes in the industries. Therefore, we believe that when knowledge workers have a strong awareness of AI, they will apply this technology promptly to optimise their work content and

work structure thereby achieving the goal of improving their work efficiency and working ability, which promotes PSE.

SET states that when individuals have a strong sense of self-efficacy, they make positive assumptions about whether their behaviour will achieve good outcomes and they engage in more social activities to obtain more resources (Ajzen 2002); conversely, when individuals have weak sense of self-efficacy, they are more inclined to make negative assumptions, and they minimise nonessential social activities to avoid wasting resources (Zimmerman 2000). Although OCB is considered an altruistic behaviour that will not be directly rewarded by the organisation, employees are still rewarded in other senses for performing OCB. Such rewards may be on a practical or psychological level and may be short-term or long-term. When employees have a strong sense of self-efficacy, they have a higher degree of identification with their membership in the organisation and take the initiative to implement more OCB, which not only contributes to the better development of the organisation but also helps the employees themselves to gain good interpersonal relationships in the workplace (Teng et al. 2020). Therefore, we believe that when employees' PSE is stronger, on the one hand, they will be more willing to assist their colleagues in accomplishing their job content, and on the other hand, they will be more willing to accomplish tasks that are not within their scope of responsibility, i.e. they will implement more OCB to obtain more potential rewards.

In summary, based on SET and within the framework of SOR model, we consider PSE as an important influence path in the relationship between AI awareness and knowledge workers' OCB. Knowledge workers with higher AI awareness can make better use of this new technology to help them increase their work efficiency thus allowing them to have more energy to accomplish more work, which enhances their PSE. Further, employees' subjective willingness to act altruistically is elevated, which increases OCB. Therefore, this study concludes that for knowledge workers, PSE is a significant path of influence between AI awareness and OCB. We propose the following hypothesis:

H1. AI awareness promotes OCB by positively influencing PSE.

2.5. The mediating role of CSE

CSE refers to individuals' confidence in their ability to be creative in accomplishing a given task (Tierney and Farmer 2002). Existing research has found that when

individuals have low CSE it can lead to negative emotions such as anxiety in employees (Gong, Huang, and Farh 2009). Individuals with higher levels of trust in AI will rely more on AI (Pelau, Dabija, and Stanescu 2024), for knowledge workers with higher AI awareness, they are more willing to apply this new technology in their work scenarios. However, in recent years, scholars have found that AI reduces employees' ability to be creative and that employees' over-reliance on AI to complete their work tasks can lead to the fact that they will gradually lose their creative thinking (Braganza et al. 2021). They only need to follow the preset algorithms to complete their work without relying on their independent thinking ability to solve problems, and this reliance on AI technology will make the originally challenging work boring so that they gradually lose the ability to think deeply about the problem, which will lead to the inhibition of creative solutions, and their self-efficacy will be reduced in the process (Strich, Mayer, and Fiedler 2021). In addition, knowledge workers tend to work in complex, uncertain patterns and rely on their individual talent and creativity to drive important work outcomes (Drucker 1999). Compared to other types of employees, they place a higher value on their ability to innovate (Shujahat et al. 2019) and when their ability to innovate declines, their CSE decreases significantly. Thus, for knowledge workers, AI awareness has a negative impact on CSE.

Consistent with the previous description of the application of SET in this scenario, when employees' CSE decreases, they lose confidence in completing creative tasks, which are critical parts of the knowledge worker's daily work portion, and their behaviours are affected by this (Tierney and Farmer 2011), and they will tend to reduce the implementation of the OCB to protect their resources within the workplace to counteract their perception of the uncertainty caused by the decline in their working ability.

In summary, based on SET and within the framework of SOR model, this study suggests that CSE is another important mechanism by which AI awareness affects OCB. When knowledge workers have higher AI awareness, they may have a stronger dependence on AI technology, and this dependence may reduce their creative behaviours at work thus leading to a decrease in their CSE. At this time, they may doubt their working ability, which may lead to a decrease in OCB. Therefore, our study proposes that CSE is an important path between AI awareness and OCB for knowledge workers. We propose the following hypothesis:

H2. AI awareness inhabits OCB by negatively influencing CSE.

2.6. The moderating role of OIC

Current academics have different definitions of the concept of OIC. Some scholars believe that OIC is the subjective perception of employees based on organisational scenarios (Isaksen and Akkermans 2011; Shanker et al. 2017), while other scholars believe that OIC reflects the inherent characteristics of the organisation and is a thing that exists objectively in reality (Hsu and Chen 2017). It is necessary to adopt appropriate definitions for research due to different needs. For this study, it is more accurate to study the influence of the environment on individual behaviour from the individual's subjective perception, so our study considers OIC as the degree to which members perceive their organisation's support for their innovative behaviour (Malibari and Bajaba 2022). At a time when AI is a technology that is just beginning to be widely marketed and applied, the OIC has a key influence on individuals' acceptance of emerging technologies (Sahoo et al. 2024; Zeng et al. 2024). The higher the perceived OIC, the stronger the individual's acceptance of the potential failure of innovative behaviours, and therefore the stronger the individual's acceptance of new technologies, and the more they are willing to try new technologies in work situations. The lower the perceived OIC, the weaker the individual's receptivity to the potential failure possibility of innovative behaviour, and therefore the weaker the individual's receptivity to new technologies, and they are more inclined to traditional and conservative production and work methods (Kong et al. 2024). Based on this, it can be inferred that the OIC also affects the use of AI in the organisation.

Existing studies have concluded that at the macro level, a good OIC helps to develop new technologies and apply them in production, so that organisational resources can be allocated more rationally (Liu and Chan 2017). At the micro level, a high level of OIC encourages employees to actively try to learn new technologies and improves their work enthusiasm and stress resistance (Lv et al. 2021). Overall, a good OIC contributes to both organisational performance and individual employee performance. For knowledge workers, when they perceive a higher degree of OIC, they will be more willing to believe that the new technology can improve the status quo and have confidence in mastering the new technology, and at this time their willingness to learn and apply the new technology will be stronger (Liang et al. 2022). In the context of the current blowout development of AI technology, knowledge workers who perceive a positive OIC are more willing to take the initiative to learn and try to master AI technology, and their work performance will benefit from

the increased productivity of AI technology as a result (Prentice and Nguyen 2020). According to SET, people usually internalise the environmental changes that lead to their job performance as subjective feelings of their competence enhancement, and this perceived competence experienced because of performance success changes their cognition and allows them to make positive associations with unknown outcomes that may result from the behaviour in question, i.e. the employees' PSE self-efficacy is enhanced. Based on this, this study speculates that OIC can promote the positive impact of AI awareness on employees' PSE.

OIC can motivate knowledge workers to be proactive in the process of innovation (Jo 2023). As previously elaborated, the application of AI may negatively affect the CSE of knowledge workers. However, a positive OIC promotes knowledge sharing and teamwork within the organisation, which facilitates the sharing of AI technology applications among employees, pushing knowledge workers to apply this technology to different work scenarios proactively. In this process, we infer that with the support of a good OIC, knowledge workers' application and innovation of AI will mitigate the negative impact of the advancement of AI technology on their independent innovation (Malik, Nguyen, and Budhwar 2024). In addition, a good OIC encourages employees to invest the idle resources generated by AI technology into more innovative behaviours that benefit their CSE, which also mitigates the negative impact of AI awareness on knowledge workers' CSE. Based on this, our study speculated that OIC can mitigate the negative impact of AI awareness on employees' CSE.

Based on the above analysis we propose the following hypotheses:

H3a. OIC positively moderates the effect of AI awareness on PSE.

H3b. OIC negatively moderates the effect of AI awareness on CSE.

Furthermore, PSE and CSE are two mediating paths of the effect of AI awareness on OCB, and we argue that OIC also moderates the indirect effects through PSE and CSE. Based on H1 and H3a, when knowledge workers' subjective perception of OIC is higher, employees with higher AI awareness can apply new technologies to their work to a greater extent, and their productivity is enhanced. At this time, PSE increases significantly, which contributes to the increase in OCB. In other words, the stronger the OIC, the stronger the positive indirect effect of PSE between AI awareness and OCB. Based on H2 and H3b, when knowledge workers' subjective perception of OIC is higher, the negative effect of AI awareness on CSE is mitigated, and relatively higher CSE increases OCB. In other words, the stronger the OIC is, the weaker the negative indirect effect of CSE is between AI awareness and OCB.

Therefore, we further propose the following hypotheses:

H4a. OIC positively moderates the indirect effect of AI awareness on OCB through PSE.

H4b. OIC negatively moderates the indirect effect of AI awareness on OCB through CSE.

The model for this study is shown in Figure 1.

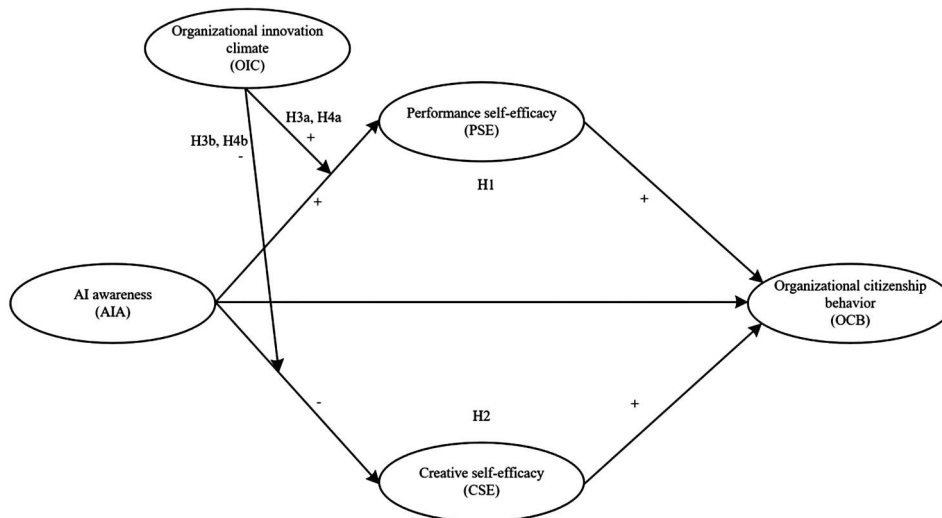


Figure 1. Model framework.

3. Method

3.1. Participants and procedures

The survey for this study was conducted from October 2023 to January 2024. Through alumni and private social relationships, we invited a total of 587 knowledge workers from a total of 11 cities in mainland China, including internet, advertising, manufacturing, banking & finance, consulting, healthcare, quality testing, and research & education industries, to participate in this study. The research was conducted with the help of HR departments as well as managers of the corresponding companies, and all the questionnaires were offline questionnaires. The specific process was as follows: first, the HR department or manager determined the list of employees who were knowledge workers and could participate, then conveyed the purpose and significance of the questionnaire to the employees, emphasised the confidentiality and anonymity of the questionnaire data to make them feel at ease to participate in the study and thus guarantee the authenticity and objectivity of the data, and finally determined the number of participants based on the principle of voluntary participation. The questionnaires were distributed at the end of the collective meetings of the respondent companies. In order to reduce the potential impact of common method bias, and with reference to the existing literature, we used a method of collecting data at different time points: in the first time, subjects were first asked to fill in questions related to demographics, AI awareness, and OCB (time point 1), and one month later, they were asked to fill in questions related to PSE, CSE, and OIC (time point 2), and data tracking matching between the two processes was implemented using anonymous coding. At time point 1, 530 questionnaires were successfully collected, and after excluding 29 invalid questionnaires with omitted or duplicated options, 501 valid questionnaires remained; at time point 2, the remaining part of the study was based on the 501 employees remaining from the previous questionnaire collection process, and the final number of valid questionnaires remaining after excluding 40 invalid questionnaires was 461, with a response rate of 86.98%, which is considered acceptable based on the experience of previous studies (Cao, Yan, and Teng 2023).

The demographics of the study are shown in Table 1. In terms of gender, 42.08% are female and 57.92% are male. In terms of age, 14.97% of the employees were below 24 years old, 31.02% were between 25–34 years old, 37.09% were between 35–44 years old, 10.85% were between 45–54 years old, and 6.07% were 55 years old and above. In terms of education level, the collected data fully proved the effectiveness of the

Table 1. Demographic information.

Variable	Category	Frequency	%
Gender	Female	194	42.08
	Male	267	57.92
Age	24 or below	69	14.97
	25–34	143	31.02
	35–44	171	37.09
	45–54	50	10.85
	55 or above	28	6.07
Education	Secondary school or below	32	6.94
	Bachelor's degree	203	44.03
	Master's degree	143	31.02
	Doctoral degree	83	18.01
Organisation tenure (year)	Less than 1	69	14.97
	1–5	157	34.06
	6–10	111	24.08
	11–15	87	18.87
	16 or above	37	8.02

Note: n = 461.

screening of knowledge workers, with 6.94% of all subjects with secondary school or below, 44.03% with a bachelor's degree, 31.02% with a master's degree, and 18.01% with a doctoral degree. In terms of organisational tenure, the percentage of less than 1 year was 14.97%; 1–5 years was 34.06%; 6–10 years was 24.08%, 11–15 years was 18.87%, and 16 years and above was 8.02%.

3.2. Measures

All question items were measured by using a 5-point Likert scale, where 1 indicates strongly disagree and 5 indicates strongly agree. All the selected scales are English scales which are widely used in the literature in authoritative journals. Since the research scenario was in mainland China, we used the back-translation method to translate all the scales (Brislin 1970).

Due to the tremendous development and advancement of AI technology within the last few years, the choice of the AI awareness scale was made by discarding the more time-honoured and mature scale and instead using the 4-item scale from Liang et al. (2022) to measure AI awareness. An example item is: 'I think there is a possibility that my current job will be replaced by AI'. With reference to existing research (Tang and Wei 2022), we used a 4-item scale from Hirst et al. (2018) to measure PSE, with the example item: 'I have a knack for using existing knowledge to improve work practices and processes'. For the measurement of CSE, we used the scale developed by Tierney and Farmer (2002) which consists of three items, with the example item: 'I feel I am good at generating novel ideas'. For the measurement of OCB, we used the 16-item scale from Lee and Allen (2002). An example item is: 'Take action to protect the organization from potential problems'. We used the scale from Ye, Liu, and Tan

(2022) to measure OIC. The scale has 4 items, an example item is 'In my organization, new ideas can easily get support and help'. Referring to the existing literature, we selected age, education and organisational tenure as control variables thereby reducing interference (Darvishmotevali and Altinay 2022; Ma, Teng, and Yan 2024).

4. Results and data analysis

Before formally conducting the analysis, we first considered the normality of the data. The results of descriptive statistics are shown in Table 2, where the absolute values of skewness and kurtosis for each construct reach the decision intervals of ± 2 and ± 7 (Curran, West, and Finch 1996). We consider the data to have good normality.

4.1. Common method bias

We used Harman's one-factor test to measure the common method bias, and the result was 35.91%, which was better than the critical criterion of 50% (Podsakoff et al. 2003). In addition, referring to existing studies (Cao, Yan, and Teng 2023), comparing the models with different numbers of factors found that Model 1 (5 factors model) had the optimal fit indices (see Table 3). Therefore, there is no potential common method bias problem in the model.

4.2. Measurement model

The model fit indices for this study: $\chi^2/df = 4.423$, CFI = 0.893, TLI = 0.882, NFI = 0.866, IFI = 0.890, GFI = 0.865, RMSEA = 0.087, and SRMR = 0.043. The

cross loadings between the different constructs are shown in Table 4, and each item was found to be at its own construct maximised and all were greater than the 0.1 criterion (Gefen and Straub 2005). These results provide support for the good fit of the model and the good setting up of the questionnaire.

4.3. Reliability and validity

As shown in Table 5, the Cronbach's α and CR values of all variables and loading values of all items are greater than 0.7 (Churchill 1979; Raza and Khan 2022), which indicates that the questionnaire has good reliability. The AVE values for all variables are greater than 0.5, which indicates good convergent validity (Hair et al. 2017). As shown in Table 6, the square root of the AVE of each variable in the correlation matrix reached its maximum value, and as shown in Table 7, all hetero-trait-monotrait ratios reached the requirement of less than 0.85 (Franke and Sarstedt 2019), and combined with the previous analysis of the results in Table 3, shows that the discriminant validity is good. Therefore, the reliability and validity of the model were considered good.

4.4. Hypotheses test

The test results of the model including direct effects, indirect effects, and moderating effects are shown in Table 8.

4.4.1. Direct effects test

The path coefficient of AIA to OCB is 0.409 ($t = 9.613$, $p < 0.001$; 95% confidence interval (CI) = [0.366, 0.553]), indicating that AI awareness has a significant positive

Table 2. Descriptive statistics.

	N	Mean	SD	Median	Min	Max	Skewness	Kurtosis
AIA	461	3.68	0.84	4.00	1.00	5.00	-0.77	0.51
PSE	461	3.28	0.97	3.25	1.00	5.00	-0.06	-0.80
CSE	461	3.64	0.84	4.00	1.00	5.00	-0.67	0.41
OIC	461	2.50	1.10	2.00	1.00	5.00	0.77	-0.29
OCB	461	3.19	0.93	3.44	1.44	5.00	-0.43	-1.03

Table 3. Model fit comparison.

	χ^2/df	CFI	TLI	NFI	IFI	GFI	AGFI	RMSEA	SRMR
Model 1	3.573	0.905	0.896	0.874	0.906	0.802	0.769	0.075	0.04
Model 2	5.291	0.84	0.827	0.811	0.841	0.724	0.68	0.096	0.077
Model 3	6.938	0.778	0.76	0.75	0.778	0.669	0.619	0.113	0.102
Model 4	9.186	0.692	0.669	0.668	0.693	0.594	0.535	0.133	0.118
Model 5	12.606	0.562	0.531	0.543	0.564	0.519	0.45	0.159	0.145

Model 1: AIA, PSE, CSE, OCB, OIC.

Model 2: AIA + PSE, CSE, OCB, OIC.

Model 3: AIA + PSE + CSE, OCB, OIC.

Model 4: AIA + PSE + CSE + OCB, OIC.

Model 5: AIA + PSE + CSE + OCB + OIC.

Table 4. Cross loading.

	AIA	CSE	OCB	OIC	PSE
AIA1	0.892	−0.256	0.334	−0.135	0.180
AIA2	0.881	−0.195	0.307	−0.121	0.144
AIA3	0.882	−0.186	0.330	−0.099	0.171
AIA4	0.864	−0.133	0.355	−0.081	0.186
CSE1	−0.155	0.929	0.233	−0.101	−0.036
CSE2	−0.238	0.904	0.138	−0.078	−0.103
CSE3	−0.202	0.858	0.111	−0.074	−0.073
OCB1	0.263	0.175	0.744	0.004	0.165
OCB10	0.297	0.091	0.796	0.012	0.127
OCB11	0.315	0.132	0.828	0.051	0.109
OCB12	0.342	0.119	0.840	0.031	0.166
OCB13	0.336	0.153	0.858	0.023	0.157
OCB14	0.320	0.170	0.855	0.005	0.208
OCB15	0.298	0.155	0.838	−0.004	0.168
OCB16	0.296	0.209	0.850	0.013	0.190
OCB2	0.313	0.207	0.815	−0.03	0.168
OCB3	0.327	0.153	0.821	0.050	0.187
OCB4	0.320	0.136	0.811	0.044	0.186
OCB5	0.324	0.144	0.764	−0.040	0.191
OCB6	0.301	0.130	0.787	0.006	0.140
OCB7	0.277	0.161	0.821	0.019	0.149
OCB8	0.266	0.102	0.806	0.044	0.115
OCB9	0.333	0.118	0.869	0.053	0.169
OIC1	−0.107	−0.096	0.033	0.935	0.041
OIC2	−0.106	−0.027	0.013	0.862	0.015
OIC3	−0.163	−0.024	0.015	0.871	0.010
OIC4	−0.111	−0.109	0.008	0.930	−0.012
PSE1	0.141	−0.152	0.153	0.024	0.837
PSE2	0.178	−0.081	0.154	−0.015	0.835
PSE3	0.150	−0.011	0.132	0.029	0.791
PSE4	0.173	−0.025	0.213	0.014	0.858

Table 5. Reliability and validity.

Constructs	Items	Loading	Cronbach's α	CR	AVE
AIA	AIA1	0.892	0.903	0.932	0.774
	AIA2	0.881			
	AIA3	0.882			
	AIA4	0.864			
CSE	CSE1	0.929	0.879	0.926	0.806
	CSE2	0.904			
	CSE3	0.858			
	CSE4	0.869			
OCB	OCB1	0.744	0.967	0.970	0.672
	OCB10	0.796			
	OCB11	0.828			
	OCB12	0.840			
	OCB13	0.858			
	OCB14	0.855			
	OCB15	0.838			
	OCB16	0.850			
	OCB2	0.815			
	OCB3	0.821			
	OCB4	0.811			
	OCB5	0.764			
OIC	OCB6	0.787			
	OCB7	0.821			
	OCB8	0.806			
	OCB9	0.869			
	OIC1	0.935	0.931	0.945	0.811
PSE	OIC2	0.862			
	OIC3	0.871			
	OIC4	0.930			
	PSE1	0.837	0.851	0.899	0.690
	PSE2	0.835			
	PSE3	0.791			
	PSE4	0.858			

effect on OCB. The path coefficient of AIA to PSE is 0.192 ($t = 4.209$, $p < 0.001$; 95% CI = [0.119, 0.217]), indicating that AI awareness has a significant positive

Table 6. Correlation matrix.

	AIA	CSE	OCB	OIC	PSE
AIA	0.880				
CSE	−0.220	0.898			
OCB	0.377	0.182	0.820		
OIC	−0.124	−0.095	0.021	0.900	
PSE	0.194	−0.078	0.200	0.015	0.831

Note: The diagonal numbers (bold values) are the square root of each variable AVE.

Table 7. Heterotrait-monotrait ratio.

	AIA	CSE	OCB	OIC	PSE
AIA					
CSE	0.246				
OCB	0.402	0.192			
OIC	0.146	0.077	0.036		
PSE	0.219	0.102	0.214	0.036	

Table 8. Hypotheses test.

	β	S.E.	t	p	LLCI	ULCI
Direct effects:						
OCB $\leftarrow$ AIA	0.409	0.048	9.613	***	0.366	0.553
PSE $\leftarrow$ AIA	0.192	0.053	4.209	***	0.119	0.327
OCB $\leftarrow$ PSE	0.139	0.040	3.337	***	0.055	0.213
CSE $\leftarrow$ AIA	−0.218	0.048	−4.796	***	−0.324	−0.136
OCB $\leftarrow$ CSE	0.277	0.044	6.618	***	0.207	0.381
Indirect effects:						
OCB $\leftarrow$ PSE $\leftarrow$ AIA	0.027	0.011	2.615	**	0.008	0.052
OCB $\leftarrow$ CSE $\leftarrow$ AIA	−0.060	0.017	−3.884	***	−0.102	−0.034
Total indirect effects	0.376	0.048	8.701	***	0.327	0.517
Moderating effects:						
OIC $\times$ AIA $\rightarrow$ CSE	0.022	0.049	0.447	0.655	−0.072	0.121
OIC $\times$ AIA $\rightarrow$ PSE	0.183	0.052	3.492	***	0.082	0.285

Note: ** $p < 0.01$; *** $p < 0.001$.

effect on PSE. The path coefficient of PSE to OCB is 0.139 ($t = 3.337$, $p < 0.001$; 95% CI = [0.055, 0.213]), indicating that PSE has a significant positive effect on OCB. The path coefficient of AIA to CSE is −0.218 ($t = −4.796$, $p < 0.001$; 95% CI = [−0.324, −0.136]), indicating that AI awareness has a significant positive effect on CSE. The path coefficient of CSE to OCB is 0.277 ($t = 6.618$, $p < 0.001$; 95% CI = [0.207, 0.381]), indicating that CSE has a significant positive effect on OCB.

4.4.2. Indirect effects test

The path coefficient for AIA $\rightarrow$ PSE $\rightarrow$ OCB is 0.027 ($t = 2.615$, $p < 0.01$; 95% CI = [0.008, 0.052]), indicating that PSE has a significant mediating effect in the promotion of OCB by AI awareness. The path coefficient for AIA $\rightarrow$ CSE $\rightarrow$ OCB is −0.060 ($t = −3.884$, $p < 0.001$; 95% CI = [−0.102, −0.034]), indicating that CSE has a significant mediating effect in the inhibition of OCB by AI awareness.

4.4.3. Moderating effects test

The path coefficient of the interaction term of OIC and AIA affecting PSE is 0.183 ($t = 3.492$, $p < 0.001$; 95% CI = [0.082, 0.285]), indicating that OIC has a significant positive moderating effect in the process of AI awareness promoting PSE (see Figure 2). The path coefficient of the interaction term of OIC and AIA affecting CSE is 0.022 ($t = 0.447$, $p = 0.655$; 95% CI = [-0.072, 0.121]), indicating that OIC does not have a significant moderating effect in the process of AI awareness inhibiting CSE (see Figure 3).

In summary, the original hypotheses H1, H2, H3a, and H4a were supported, while H3b and H4b were not supported.

5. Discussion and conclusion

5.1. General findings

The purpose of this study is to examine the double-edged sword effect of knowledge workers' AI awareness on OCB through PSE and CSE as two mediating

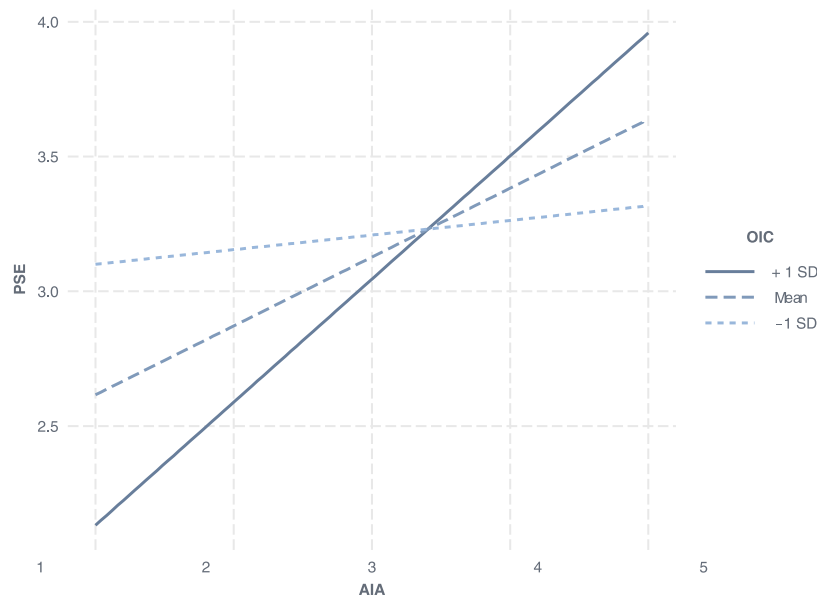


Figure 2. The moderating effect test of OIC between AIA and PSE.

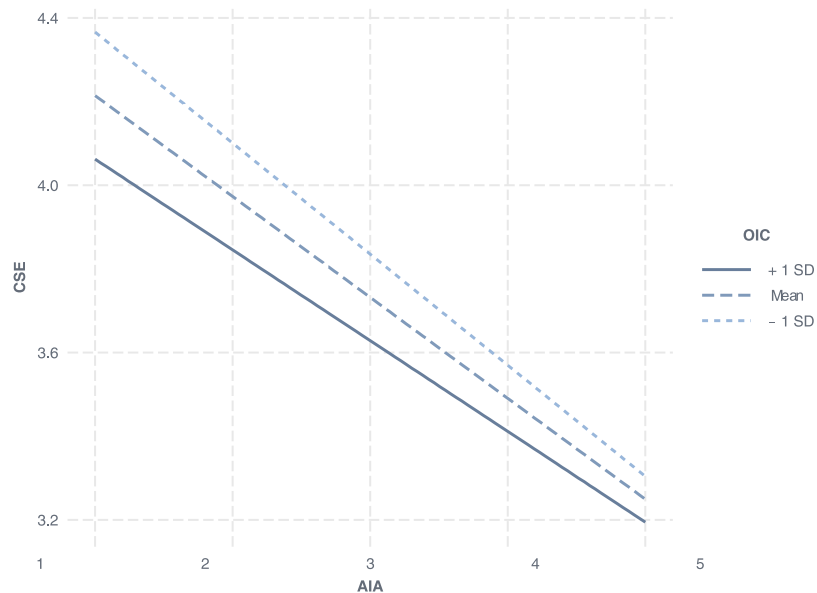


Figure 3. The moderating effect test of OIC between AIA and CSE.

variables, as well as the moderating role of OIC. The results of this research support part of the hypotheses, as shown in Figure 4 for the specific effect values and the significances.

A summarising conclusion is that AI awareness of the knowledge worker has a double-edged effect on OCB. Overall, the impact of AI awareness on OCB is positive, which is consistent in direction with the findings of previous studies, for example, some of which have pointed out that new technologies such as AI and intelligent robots reduce the workload of employees in their daily work, promote work autonomy (Zhang et al. 2023), which can help them implement more positive behaviours in their work situations (De Spiegelaere et al. 2014). However, our study equally emphasises the existence of the negative impact of AI awareness on OCB. Due to the potential threat of AI, the excellent performance of this emerging technology in handling many job components can make knowledge workers doubt their ability, which further supports previous research that AI technology can lead to negative emotions and perceptions such as anxiety, job insecurity, and so on (He, Teng, and Song 2024; Song et al. 2024). When individuals have negative emotions and perceptions, they minimise social activities thereby reducing the drain on their efficacy. Although AI awareness promotes knowledge workers' OCB in terms of outcome, this result is a combination of both positive and negative effects superimposed on each other, with the positive effect being greater than the negative effect.

Our study identified PSE and CSE as the two pathways through which knowledge workers' AI awareness had a double-edged sword effect on OCB. On the one hand, AI awareness promotes knowledge workers' PSE

and thus their OCB. This finding further confirms the existing literature that 'employee performance positively affects their self-efficacy and leads to more OCB' (Sun, Lee, and Yun 2023; Sutha 2024), and that employees with higher AI awareness are able to apply this new technology more efficiently to enhance their performance, which in turn promotes positive perceptions and outcomes. On the other hand, AI awareness inhibits knowledge workers' CSE and thus their OCB. Individuals' over-reliance on AI can lead to a decline in their ability to be creative. Although AI works by mimicking the thinking logic of the human brain, the act of innovation is very much in need of a flash of insight in orderly work, and AI is unable to perform innovative behaviours completely autonomously. In other words, knowledge workers with higher levels of AI awareness will be less creative due to their dependence on this technology, which will lead to a decrease in their CSE, thus negatively affecting the OCB, which is in line with the SET mentioned above (Ajzen 2002).

The findings partially support our hypotheses on the moderating effect of OIC, confirming the moderating effect of OIC in the process of AI awareness promoting OCB through the pathway of PSE. In organisations with stronger innovation climate, organisational members have higher levels of self-learning awareness (Kandemir and Hult 2005), and at the same time, employees in such organisations are usually willing to engage in knowledge sharing or innovation competition (Azeem et al. 2021), that is to say, after generating stronger AI awareness, knowledge workers will master this new technology as soon as possible by engaging in more self-learning and applying it to their work, which pushes their work efficiency, and in turn leading to an increase in their

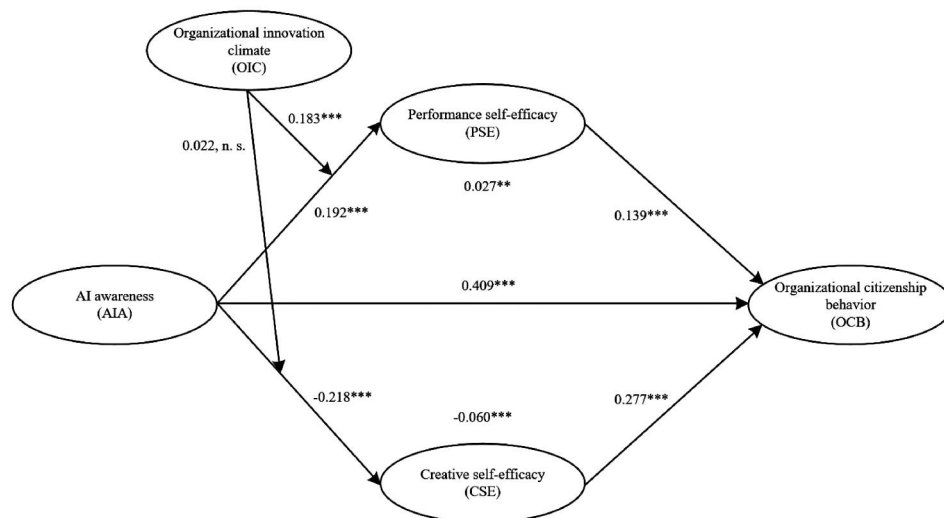


Figure 4. The result of the model.

PSE and OCB. On the other hand, the moderating effect of OIC on the process of AI awareness in inhibiting OCB through the path of CSE is not significant. The reason may lie in the fact that each knowledge worker is an independent individual, and whether or not other co-workers in the organisation pursue innovation is not relevant to one's own sense of achievement on the path of pursuing innovation, and thus whether or not co-workers around the knowledge workers are willing to innovate or not does not affect the decline of their own CSE as a result of AI awareness. In addition, CSE is more dependent on whether knowledge workers' daily work gives them enough space to perform and thus allow innovation to happen in a real sense (Jaiswal and Dhar 2015), and when it is reduced due to AI awareness, we believe that a stronger OIC drives knowledge workers' enthusiasm for innovation rather than their CSE, which is consistent in direction with existing study (Yu, Ayub, and Bashir 2024).

5.2. Theoretical implications

First, this study provides a useful addition to the field of research on the impact of AI awareness on employee behaviour. The existing literature has mainly explored the research on the impact of AI awareness on employees' in-role-related variables, such as innovative behaviour, turnover, and work performance, and our study expands for the first time the research on the impact of AI awareness on employees' out-of-role behaviours. Based on this and considering the inherent characteristics of knowledge workers, we strongly believe that compared with other types of employees, knowledge workers will have a stronger awareness and usage of AI due to their job requirements, thus making it more valuable to study. To a certain extent, our study fills the gap in the research direction of knowledge workers, which is more closely linked to AI technology.

Second, in order to explore the path of the influence of AI awareness on OCB, this study introduces PSE and creativity self-efficacy as mediating variables in both positive and negative directions based on SET. This is the first time that SET has been applied to a research scenario in the emerging popular field of AI. This study expands the application scenario of SET while opening the black box of the influence of AI awareness on employees' OCB. In addition, this study expands the application scenarios of SOR model in the field of employee behaviour.

Third, we introduced OIC to explain the moderating role of organisational factors on the main effects in the model which broadened the research perspective. This study argues that organisational-level factors cannot

be ignored when assessing the impact of a variable within an organisation on an individual, and our results partially support the original hypotheses by finding that the OIC has a gainful effect on knowledge workers' AI awareness through PSE promoting OCB.

Overall, by assessing the double-edged sword effect of AI awareness on employees' OCB by using two types of employee perception-level factors, PSE and CSE as mediating variables, and the organisational-level factor OIC as moderating variables, our study provides significant evidence that AI awareness can affect individual behaviour in both positive and negative directions, respectively. The results of the study provide support for our understanding that AI awareness has a bidirectional impact, suggesting that in addition to focusing on individual employee factors, organisational factors should not be neglected at the same time in theoretical studies.

5.3. Practical implications

First, managers must clearly realise that AI in the era of Industry 4.0 does not necessarily bring only positive or negative impacts, and there is a double-edged sword effect on organisational development. At the micro level, existing research has more often suggested that AI awareness will hurt employee behaviour, and these findings may cause anxiety and worry for managers in real-world scenarios. Our results can help managers understand the positive effects of AI awareness on the OCB of knowledge workers and alleviate such anxiety, so that managers can realise the positive aspects of AI and maximise such positive effects through management tools, thus helping organisations enjoy the benefits of technological development to the greatest extent as possible.

Second, for organisations that focus more on the positive effects of AI, this study suggests that they must be aware of the potential drawbacks, that is while recognising the positive effects of AI awareness on employees' PSE, managers need to be aware of its inhibitory effect on CSE and thus its negative impact on employee behaviour. Therefore, on the one hand, organisations should increase the recruitment of knowledge workers with higher AI awareness and improve the AI awareness of knowledge workers through training, to ensure that members of the organisation can effectively apply AI technology in appropriate scenarios to improve their PSE, which will, in turn, motivate them to implement more OCB. On the other hand, managers should give more guidance to employees with higher AI awareness in their daily work to help them correctly recognise the boundaries of the usage of AI and to

avoid the decline of creative behaviours caused by employees' over-reliance on AI, which in turn causes the decline of CSE. At the same time, given that self-efficacy is a kind of perception before the implementation of individual behaviours, we believe that managers should give employees positive feedback on their work results promptly and directly improve their PSE and CSE through encouragement, thus encouraging them to implement more OCB, which in turn benefits the organisation.

Third, managers must pay attention to the innovation climate of the organisation, which is crucial for knowledge workers with high AI awareness to implement positive behaviours. Nowadays, AI technology has become inextricably linked to the working scenarios of knowledge workers, and managers should ensure that the organisation is in a good innovation climate through management strategies, to amplify the positive impact of AI awareness on the OCB of knowledge workers. In terms of specific strategies, managers should create a positive organisational climate for innovation by including providing rewards for knowledge workers' innovative behaviours, tolerating failed innovations, and encouraging employees to connect with the outside world thereby increasing their sources of creativity. This study reveals an important link between knowledge workers' AI awareness and their OCB and guides managers on how to help organisations enjoy technological benefits in the new period of alternating technological change at the micro level of employee behaviour.

5.4. Limitations and further research

This study empirically employs cross-sectional data to validate the theoretical model based on previous experiences in the literature, which makes the application of the findings in real-life scenarios with the development of AI technology time-limited. Although AI technology has penetrated into the working scenarios of knowledge workers, with the development and application of AI technology, the AI awareness of employees may change significantly, therefore, we call on future scholars to conduct more research in the direction of AI awareness of employees affecting their behaviours, and compare with our findings, so as to timely capture the changes of employees' AI awareness affecting their behaviour.

Since this is the first time the relationship between employees' AI awareness and OCB has been established, we have been very conservative in the selection and treatment of the dependent variables by considering the different types of OCB as a whole. In fact, this is consistent with the developmental history of other research

directions for employees' OCB, and it is hoped that this study will serve as a foundational insight for subsequent studies that delve deeper into the relationship between them. In future research, scholars can refine OCB according to the implementation of different targets to conduct more targeted research.

The moderating effect of the OIC has been partially verified. In further research, scholars can try to explore the boundary conditions of the model from more dimensions to broaden the research perspectives and increase the scope of application of the model in practical scenarios. Meanwhile, although we creatively tried to use SET to explore the effect of AI awareness on OCB, the research on the effect of AI awareness on employee behaviour at the micro level is still in its infancy, and future studies can try to explore this topic more based on SET.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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